

Movie Ticket Booking System — Class Based

Movie and Booking structures are inside the class, public: is directly below them, all functions are public, and the complete 11-option menu is inside main().

```
#include <iostream.h>
#include <conio.h>
#include <stdio.h>
#include <string.h>

class MovieBooking
{
    struct Movie
    {
        int id;
        char name[40];
        char time[15];
        int price;
        int seats;
    };

    struct Booking
    {
        int ticketNo;
        char customerName[40];
        char movieName[40];
        int seats;
        int amount;
    };

public:

    Movie movies[15];
    Booking bookings[50];
    int bookingCount;

    MovieBooking()
    {
        bookingCount = 0;

        movies[0] = {1, "Avengers", "10:00 AM", 200, 50};
        movies[1] = {2, "Avatar", "11:00 AM", 180, 50};
        movies[2] = {3, "Spider-Man", "12:00 PM", 150, 50};
        movies[3] = {4, "RRR", "01:00 PM", 220, 50};
        movies[4] = {5, "Titanic", "02:00 PM", 200, 50};
        movies[5] = {6, "Chichore", "03:00 PM", 180, 50};
        movies[6] = {7, "3 Idiots", "04:00 PM", 150, 50};
        movies[7] = {8, "Bahubali", "05:00 PM", 200, 50};
        movies[8] = {9, "Pushpa", "06:00 PM", 180, 50};
        movies[9] = {10, "Animal", "07:00 PM", 220, 50};
        movies[10] = {11, "War", "08:00 PM", 200, 50};
        movies[11] = {12, "Singham", "09:00 PM", 170, 50};
        movies[12] = {13, "Drishyam", "10:00 AM", 150, 50};
        movies[13] = {14, "Brahmastra", "11:00 AM", 190, 50};
        movies[14] = {15, "Kabir Singh", "12:00 PM", 180, 50};
    }

    void saveBookings()
    {
        FILE *file;

        file = fopen("bookings.txt", "w");

        if (file == NULL)
        {
            cout << "\nFile cannot be opened!";
            return;
        }

        for (int i = 0; i < bookingCount; i++)
        {
            fprintf(file, "Ticket No: %d\n", bookings[i].ticketNo);
            fprintf(file, "Customer: %s\n", bookings[i].customerName);
            fprintf(file, "Movie: %s\n", bookings[i].movieName);
            fprintf(file, "Seats: %d\n", bookings[i].seats);
            fprintf(file, "Amount: Rs.%d\n", bookings[i].amount);
            fprintf(file, "-----\n");
        }

        fclose(file);
    }

    void displayMovies()
    {
        clrscr();

        cout << "\n=====";
        cout << "\n                *** NOW SHOWING ***";
        cout << "\n=====";

        cout << "\n\n";
    }
};
```

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cout << "ID    MOVIE NAME          SLOT          PRICE    VACANT";
cout << "\n-----";

for (int i = 0; i < 15; i++)
{
    cout << "\n";
    cout << movies[i].id << "    ";
    cout << movies[i].name;
    cout << "\t\t";
    cout << movies[i].time;
    cout << "\t";
    cout << "Rs." << movies[i].price;
    cout << "\t";
    cout << movies[i].seats;
}

cout << "\n-----";

getch();
}

void searchMovie()
{
    char searchName[40];
    int found = 0;

    clrscr();

    cout << "\n===== ";
    cout << "\n          *** SEARCH MOVIE ***";
    cout << "\n===== ";

    cin.get();

    cout << "\n\nEnter Movie Name: ";
    cin.getline(searchName, 40);

    for (int i = 0; i < 15; i++)
    {
        if (strcmpi(movies[i].name, searchName) == 0)
        {
            cout << "\n\n+-----+";
            cout << "\n|          MOVIE FOUND!          |";
            cout << "\n+-----+";

            cout << "\n Movie Name      : " << movies[i].name;
            cout << "\n Show Slot       : " << movies[i].time;
            cout << "\n Ticket Price    : Rs." << movies[i].price;
            cout << "\n Vacant Seats    : " << movies[i].seats;

            cout << "\n+-----+";

            found = 1;
            break;
        }
    }

    if (found == 0)
        cout << "\n\n*** MOVIE NOT FOUND! ***";

    getch();
}

void bookTicket()
{
    char movieName[40];
    int seats;
    int found = 0;

    clrscr();

    cout << "\n===== ";
    cout << "\n          *** BOOK TICKET ***";
    cout << "\n===== ";

    if (bookingCount < 50)
    {
        cin.get();

        cout << "\n\nEnter Customer Name: ";
        cin.getline(bookings[bookingCount].customerName, 40);

        cout << "Enter Movie Name: ";
        cin.getline(movieName, 40);

        for (int i = 0; i < 15; i++)
        {
            if (strcmpi(movies[i].name, movieName) == 0)
            {
                found = 1;

                cout << "\nMovie Found!";
                cout << "\nShow Slot : " << movies[i].time;
                cout << "\nTicket Price : Rs." << movies[i].price;
                cout << "\nVacant Seats : " << movies[i].seats;
            }
        }
    }
}

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        cout << "\n\nEnter Number of Seats: ";
        cin >> seats;

        if (seats > 0 && seats <= movies[i].seats)
        {
            bookings[bookingCount].ticketNo =
                bookingCount + 1;

            strcpy(bookings[bookingCount].movieName,
                movies[i].name);

            bookings[bookingCount].seats = seats;

            bookings[bookingCount].amount =
                seats * movies[i].price;

            movies[i].seats =
                movies[i].seats - seats;

            cout << "\n\n*****";
            cout << "\n*          TICKET BOOKED SUCCESSFULLY!          *";
            cout << "\n*****";

            cout << "\n\n          *** TICKET ***";
            cout << "\n-----";

            cout << "\n Ticket No.      : "
                << bookings[bookingCount].ticketNo;

            cout << "\n Customer       : "
                << bookings[bookingCount].customerName;

            cout << "\n Movie          : "
                << bookings[bookingCount].movieName;

            cout << "\n Show Slot      : "
                << movies[i].time;

            cout << "\n Seats           : "
                << bookings[bookingCount].seats;

            cout << "\n Total Amount : Rs."
                << bookings[bookingCount].amount;

            cout << "\n-----";
            cout << "\n          ENJOY YOUR MOVIE!";

            bookingCount++;

            saveBookings();
        }
        else
        {
            cout << "\n\n*** NOT ENOUGH SEATS AVAILABLE! ***";
        }

        break;
    }
}

if (found == 0)
    cout << "\n\n*** MOVIE NOT FOUND! ***";
}
else
{
    cout << "\n\n*** BOOKING LIMIT REACHED! ***";
}

getch();
}

void viewAllBookings()
{
    clrscr();

    cout << "\n\n=====";
    cout << "\n          *** ALL BOOKINGS ***";
    cout << "\n\n=====";

    if (bookingCount == 0)
    {
        cout << "\n\nNo bookings available!";
    }
    else
    {
        cout << "\n\n";
        cout << "TICKET      CUSTOMER          MOVIE";
        cout << "\n\n-----";

        for (int i = 0; i < bookingCount; i++)
        {
            cout << "\n"
                << bookings[i].ticketNo
                << "\t"
                << bookings[i].customerName

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        {
            movies[j].seats =
                movies[j].seats - newSeats;

            bookings[i].seats = newSeats;

            bookings[i].amount =
                newSeats * movies[j].price;

            saveBookings();

            cout << "\n\n*** BOOKING UPDATED SUCCESSFULLY! ***";
        }
        else
        {
            cout << "\n\n*** NOT ENOUGH SEATS AVAILABLE! ***";

            movies[j].seats =
                movies[j].seats -
                bookings[i].seats;
        }

        found = 1;
    }
}

break;
}
else
{
    cout << "\n\n*** INVALID NUMBER OF SEATS! ***";
    found = 1;
}
}

}

if (found == 0)
    cout << "\n\n*** BOOKING NOT FOUND! ***";

getch();
}

void cancelTicket()
{
    char movieName[40];
    int found = 0;
    int j;

    clrscr();

    cout << "\n=====
    cout << "\n          *** CANCEL TICKET ***";
    cout << "\n=====

    cin.get();

    cout << "\n\nEnter Movie Name: ";
    cin.getline(movieName, 40);

    for (int i = 0; i < bookingCount; i++)
    {
        if (strcmpi(bookings[i].movieName, movieName) == 0)
        {
            for (j = 0; j < 15; j++)
            {
                if (strcmpi(movies[j].name,
                    bookings[i].movieName) == 0)
                {
                    movies[j].seats =
                        movies[j].seats + bookings[i].seats;
                }
            }

            for (j = i; j < bookingCount - 1; j++)
            {
                bookings[j].ticketNo =
                    bookings[j + 1].ticketNo;

                strcpy(bookings[j].customerName,
                    bookings[j + 1].customerName);

                strcpy(bookings[j].movieName,
                    bookings[j + 1].movieName);

                bookings[j].seats =
                    bookings[j + 1].seats;

                bookings[j].amount =
                    bookings[j + 1].amount;
            }

            bookingCount--;

            saveBookings();
            cout << "\n\n*****";

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        cout << "\n*          TICKET CANCELLED!          *";
        cout << "\n*****";

        found = 1;
        break;
    }
}

if (found == 0)
    cout << "\n\n*** BOOKING NOT FOUND! ***";

getch();
}

void displayAvailableSeats()
{
    clrscr();

    cout << "\n=====";
    cout << "\n          *** AVAILABLE SEATS ***";
    cout << "\n=====";

    cout << "\n\n";
    cout << "ID      MOVIE NAME          SLOT          VACANT SEATS";
    cout << "\n-----";

    for (int i = 0; i < 15; i++)
    {
        cout << "\n"
            << movies[i].id
            << "      "
            << movies[i].name
            << "\t\t"
            << movies[i].time
            << "\t"
            << movies[i].seats;
    }

    cout << "\n-----";

    getch();
}

void calculateAmount()
{
    char movieName[40];
    int seats;
    int found = 0;

    clrscr();

    cout << "\n=====";
    cout << "\n          *** CALCULATE AMOUNT ***";
    cout << "\n=====";

    cin.get();

    cout << "\n\nEnter Movie Name: ";
    cin.getline(movieName, 40);

    for (int i = 0; i < 15; i++)
    {
        if (strcmpi(movies[i].name, movieName) == 0)
        {
            found = 1;

            cout << "\nMovie Name : " << movies[i].name;
            cout << "\nTicket Price : Rs." << movies[i].price;

            cout << "\nEnter Number of Seats: ";
            cin >> seats;

            if (seats > 0 && seats <= movies[i].seats)
            {
                cout << "\nTotal Amount = Rs."
                    << seats * movies[i].price;
            }
            else
            {
                cout << "\n*** INVALID NUMBER OF SEATS! ***";
            }

            break;
        }
    }

    if (found == 0)
        cout << "\n\n*** MOVIE NOT FOUND! ***";

    getch();
}

void totalBookings()
{
    clrscr();

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        cout << "\n=====";
        cout << "\n          *** TOTAL BOOKINGS ***";
        cout << "\n=====";

        cout << "\n\nTotal Number of Bookings : "
              << bookingCount;

        getch();
    }
};

int main()
{
    MovieBooking obj;
    int choice;

    do
    {
        clrscr();

        cout << "\n";
        cout << "*****\n";
        cout << "**                               *\n";
        cout << "**          MOVIE TICKET BOOKING SYSTEM          *\n";
        cout << "**                               *\n";
        cout << "**          WELCOME TO CINEMA                      *\n";
        cout << "**                               *\n";
        cout << "*****\n";

        cout << "\n          MAIN MENU";
        cout << "\n-----";

        cout << "\n [1] Display All Movies";
        cout << "\n [2] Search Movie";
        cout << "\n [3] Book Ticket";
        cout << "\n [4] View All Bookings";
        cout << "\n [5] Search Booking";
        cout << "\n [6] Update Booking";
        cout << "\n [7] Cancel Ticket";
        cout << "\n [8] Display Available Seats";
        cout << "\n [9] Calculate Booking Amount";
        cout << "\n [10] Total Bookings";
        cout << "\n [11] Exit";

        cout << "\n-----";

        cout << "\n\nEnter your choice: ";
        cin >> choice;

        if (choice == 1)
            obj.displayMovies();

        else if (choice == 2)
            obj.searchMovie();

        else if (choice == 3)
            obj.bookTicket();

        else if (choice == 4)
            obj.viewAllBookings();

        else if (choice == 5)
            obj.searchBooking();

        else if (choice == 6)
            obj.updateBooking();

        else if (choice == 7)
            obj.cancelTicket();

        else if (choice == 8)
            obj.displayAvailableSeats();

        else if (choice == 9)
            obj.calculateAmount();

        else if (choice == 10)
            obj.totalBookings();

        else if (choice == 11)
        {
            clrscr();

            cout << "\n\n*****";
            cout << "\n* Thank you for using the system! *";
            cout << "\n*****";

            getch();
        }

        else
        {
            cout << "\n\n*** INVALID CHOICE! ***";
            getch();
        }
    }
}

```

```
    }  
    } while (choice != 11);  
    return 0;  
}
```